

Experiment No. 2

Interfacing LED and Switch with LPC2148

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GPIO Programming: Interfacing LED and Switch with LPC2148

2.1 Course Outcome Addressed

CO3: Implement ARM based LPC2148 system for on-chip and off-chip peripheral devices.

2.2 Learning Outcomes

1. Configure GPIO pins as input/output.
2. Interface LED and switch with LPC2148.
3. Develop Embedded C programs.
4. Verify GPIO operation using hardware.

2.3 Aim

To interface an LED and push button switch with LPC2148 using GPIO programming.

2.4 Objectives

- Study GPIO registers.
- Interface LED and switch.
- Control LED using switch input.

2.5 Hardware Required

- PC-256 RAM, 1.8GHZ, 40GB HDD loaded with above software.
- LPC2148 development board with programmer
- LED module and Switch

2.6 Software Required

- KEIL IDE V4.74 with ARM Development tools chain.
- ISP programmer or Flash magic.
- Proteus design suit.
- **Operating System:-** WINDOWS 98/2000/XP/2007/2010

2.7 Theory

GPIO (General Purpose Input Output) pins of LPC2148 can be configured as digital inputs or outputs.

Registers IODIR0, IOSET0, IOCLR0 and IOPIN0 are used for GPIO configuration and control.

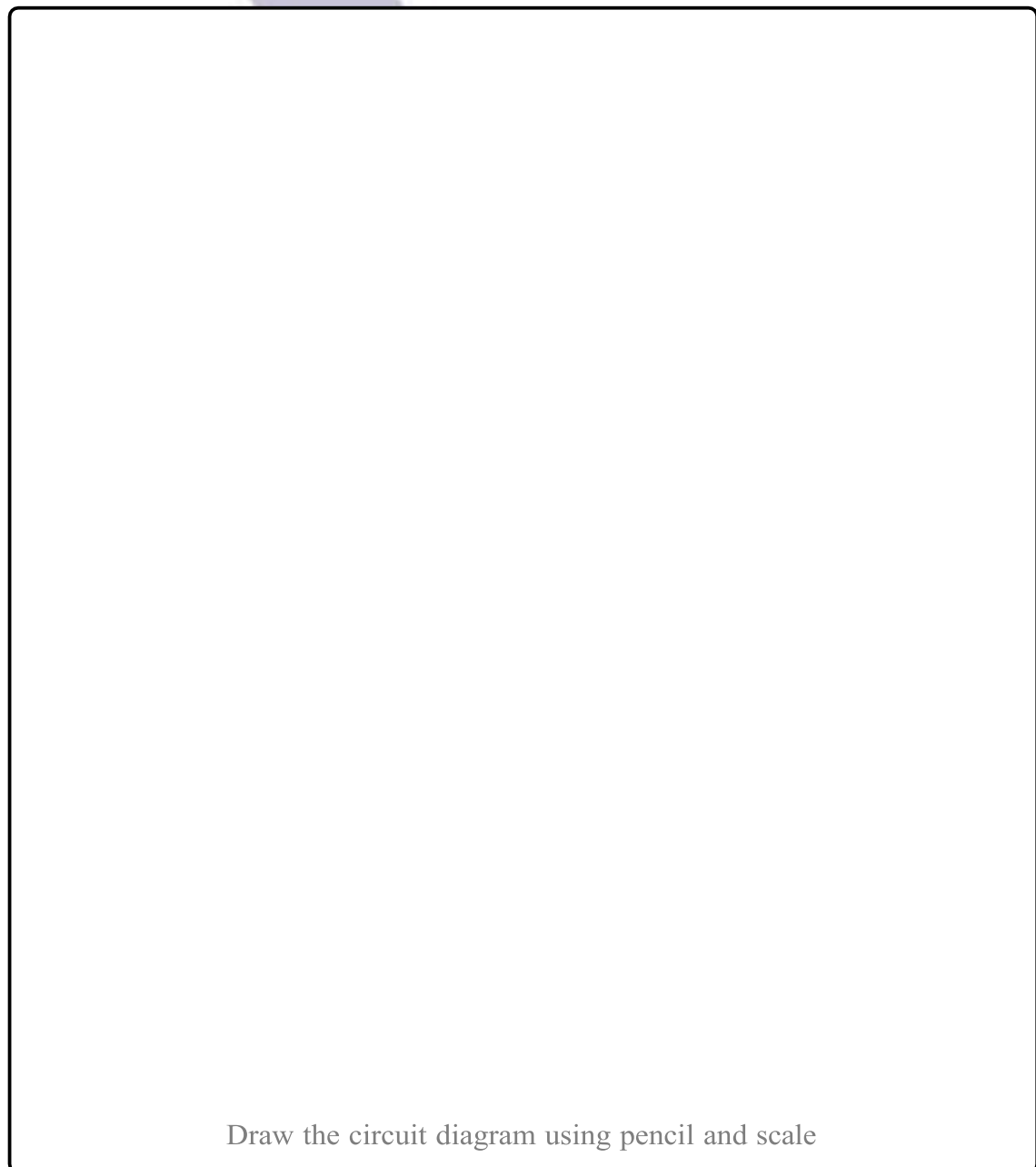
2.7.1 GPIO Registers

Sr. No.	Register	Function
1	IODIR0	Configures the GPIO pins as input or output. A bit value of '1' configures the corresponding pin as an output, while '0' configures it as an input.
2	IOSET0	Sets the selected GPIO output pins to logic HIGH without affecting the other pins.
3	IOCLR0	Clears the selected GPIO output pins to logic LOW without affecting the other pins.
4	IOPIN0	Reads the current logic level of GPIO pins and can also be used to write data to the port.

2.7.2 Pin Configuration

GPIO Pin	Interfaced Device	Direction
P0.10	LED	Output
P0.15	Push Button Switch	Input

2.8 Interfacing Diagram:



Draw the circuit diagram using pencil and scale

Algorithm: Switch Controlled LED Using GPIO

1. Start the program.
2. Include the header file LPC214x.h.
3. Configure GPIO pin P0.10 as an output by setting the corresponding bit in the IODIR0 register.
4. Configure GPIO pin P0.15 as an input by clearing the corresponding bit in the IODIR0 register.
5. Enter an infinite while(1) loop.
6. Read the logic level of the switch connected to P0.15 using the IOPIN0 register.
7. Check the status of the switch.
8. If the switch is pressed (logic LOW):
 - (a) Set P0.10 HIGH using the IOSET0 register.
 - (b) The LED turns ON.
9. Otherwise (switch is not pressed):
 - (a) Clear P0.10 using the IOCLR0 register.
 - (b) The LED turns OFF.
10. Continue monitoring the switch status by repeating Steps 6 to 9.
11. End (the program executes continuously in the infinite loop).

2.9 Result:

The LED and switch were successfully interfaced with the LPC2148 using GPIO programming. The LED turned ON when the switch was pressed and OFF when it was released, confirming the correct operation of the GPIO input and output pins.

2.10 Conclusion:

The experiment successfully demonstrated the interfacing of an LED and a switch with the LPC2148 microcontroller. It provided an understanding of GPIO programming and verified the proper operation of digital input and output using the LPC2148.

EXPERIMENT NO. 2. INTERFACING LED AND SWITCH WITH LPC2148

